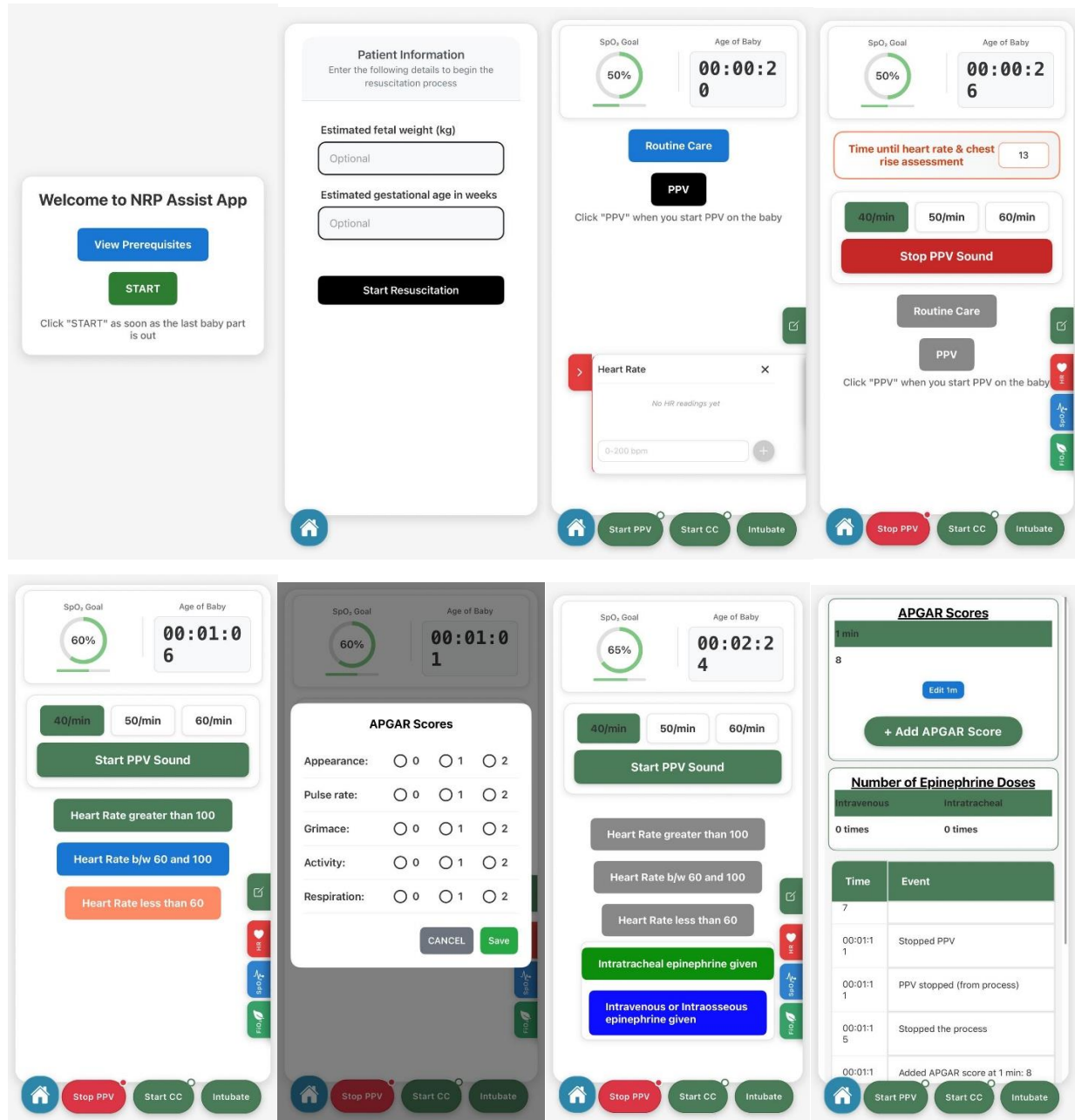


NRP Assist app Investigator Brochure:

Screenshots from the NRP Assist App:



1. Introduction

The NRP Assist application is an iOS-based decision-support tool designed to assist neonatal resuscitation providers by offering time-based prompts and reference information consistent with the Neonatal Resuscitation Program (NRP) 8th Edition guidelines¹ (2021). The app is intended to support cognitive organization and task tracking during neonatal resuscitation but is not intended to replace clinical judgment or formal NRP training.

The application is intended for use during NRP simulation and is not intended for real-life clinical use. It is essential to emphasize that this app is not a substitute for comprehensive medical training or certification in resuscitation techniques. Users must not rely on this app as their sole source of knowledge.

2. Purpose of the App

The purpose of the NRP Assist app is to support adherence to neonatal resuscitation algorithm as per the NRP 8th edition and cognitive load reduction for trained providers.

3. Key Functions

- **Timers & Alerts:** Initiates timers at key resuscitation steps and provides prompts aligned with NRP guidance.
- **Apgar Prompts:** Sends reminders and allows real-time Apgar input.
- **Oxygen Saturation Reference:** Displays expected SpO₂ values by minute according to NRP targets.
- **Epinephrine Dose Calculator:** Calculates dose based on entered estimated fetal weight.
- **Event Recording:** Generates a pdf at the end, 'Resuscitation log', with events which occurred during the resuscitation process recorded in real time.

4. User Instructions

Intended User Population:

Neonatal resuscitation providers who are trained and credentialed in NRP, including physicians, nurses, and advanced practice providers.

Basic Use Steps:

- a. Open the NRP Assist app on an iOS device (iphone or ipad)
- b. After you agree to the disclaimer, you can either view the Prerequisites or Start.
- c. An “NRP Assist App User Guide” option is available in the upper left corner of the screen and directs users to this page.
- d. Under ‘View Prerequisites’, you can see the four pre-birth questions, the NRP quick equipment checklist as well as editable team roles with 3 or 4 members present.
- e. The patient information fields including estimated fetal weight in kilograms and the estimated gestational age in weeks are optional. If you do not enter the estimated fetal weight here, you will be prompted to enter it later in the process, in order to give you the calculated epinephrine dose.
- f. Entering the estimated fetal weight and/or gestational age will generate guidance below, including intratracheal and intravenous epinephrine doses, ETT size and depth, starting FiO₂, and initial PIP and PEEP.
- g. On every page moving forward from here, you will see the age of the baby since birth on the upper left corner.
- h. Starting at 2 minutes of life, the SpO₂ goal for the infant’s age in minutes also appears on the upper left corner.
- i. APGAR score prompts appear at 1, 5, 10, 15 and 20 minutes at which point you can enter all or some of the component scores or click cancel to continue with the process
- j. The audio guidance for PPV and chest compressions will start automatically when the corresponding buttons ‘start PPV’ or ‘start CC’ are clicked. The audios can be stopped at anytime by clicking ‘stop PPV sound’ and ‘stop CC metronome’
- k. You will be prompted to assess heart rate or do certain interventions based on time since the last input into the app, by vibration as well as audio and visual prompts. The audio prompts can be stopped at any time during the process.

- l. The side pop-up buttons give you the option to enter heart rate (HR), SpO2, FiO2 and/or enter any other notes relevant at that point during the resuscitation
- m. The blue home button at the bottom left corner of the screen allows you to go back to the first page to start resuscitation with a fresh scenario
- n. When you click STOP—either after the heart rate is more than 100 bpm or after 20 minutes of resuscitation—a PDF titled *Resuscitation Log* is automatically generated. This document records all events in real time, including Apgar scores and the number of epinephrine doses administered. The PDF is editable and can be saved or shared via email or message.

5. Safety Information & Limitations

- The app is not a substitute for NRP certification or clinical expertise.
- It serves only as an adjunct tool and must not be used as the sole source of clinical decision-making.
- Presently, the application is specifically designed to be utilized exclusively within the context of NRP simulation and is not intended for real-life emergencies. It should not be used outside the context of simulation.
- The ‘resuscitation log’ generated at the end of the scenario can be stored on the work device and shared to others via email or message.

Data Privacy: The app does not collect, transmit, or store any identifiable personal data. No patient identifiers are entered except for the time of birth, estimated fetal weight and gestational age, which in this case will be of the simulation scenario.

6. Device Requirements

- Apple iOS compatible device
- Internet connection is not required for use after installation

7. Evidence & Guideline Basis

The features of the app are based on the Textbook of Neonatal Resuscitation, 8th Edition¹ and current AAP/ACOG guidelines for neonatal resuscitation and Apgar scoring.

Appendix A — Acronyms

- NRP — Neonatal Resuscitation Program
- PPV — Positive Pressure Ventilation
- CPAP — Continuous Positive Airway Pressure
- CC – Chest Compressions
- *Alternate airway* — Endotracheal intubation or laryngeal mask placement

Appendix B — APGAR Scoring Table²

Indicator	0 point	1 point	2 points
Activity (muscle tone)	Floppy	Flexed limbs	Active
Pulse rate	Absent	<100 per minute	>100 per minute
Grimace (reflex irritability)	Absent	Minimal response to stimulation	Prompt response to stimulation
Appearance (skin color)	Blue or pale	Pink body with blue extremities	Pink
Respiration	Absent	Slow and irregular	Vigorous cry

Appendix C — References

1. Textbook of Neonatal Resuscitation 8th edition
Edited by: Gary M. Weiner, MD, FAAP, Jeanette Zaichkin, RN, MN, NNP-BC
DOI: <https://doi.org/10.1542/9781610025256>
Link: <https://publications.aap.org/aapbooks/book/694/Textbook-of-Neonatal-Resuscitation?autologincheck=redirected>
2. AMERICAN ACADEMY OF PEDIATRICS COMMITTEE ON FETUS AND NEWBORN; AMERICAN COLLEGE OF OBSTETRICIANS AND GYNECOLOGISTS COMMITTEE ON OBSTETRIC PRACTICE. The Apgar Score. Pediatrics. 2015 Oct;136(4):819-22. doi: 10.1542/peds.2015-2651. PMID: 26416932.